

Exploring Fuzzy Logic and Random Forest for Car Drivers' Fuel Consumption Estimation in IoT-Enabled Serious Games

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Abstract— *Internet of Things (IoT) technologies have a promising potential for instructional serious games related to field operations. We explore IoT's potential for serious games in the automotive application domain to improve driving, choosing fuel consumption (FC) as an indicator of the driver performance as it is strongly influenced by driving styles and can be quantified and validated. We propose a FC prediction model, exploiting three vehicular signals that are controllable by the driver (player), that are able to provide direct coaching feedback to the driver and are easily available through the widely available On-Board Diagnostic-II (OBD-II) vehicular interface: throttle position, engine rotation speed (RPM) and car speed. We processed the data with two techniques, random forest (RF) and fuzzy logic (FL). Implementation, training and testing of both models, were made using the enviroCar database which freely provides a significant amount of naturalistic drive data. Results show that RF achieves quite a higher estimation accuracy, which complements FL's ability to provide driver with easily understandable feedback. We thus argue that the combination of the two models can supply valuable information usable by game designers in the automotive environment.*

Keywords — *IoT, Serious Games, user modelling, random forest, machine learning, fuzzy logic, fuel consumption, On-Board Diagnostics-II (OBD-II);*

I. INTRODUCTION

Serious Games (SGs) are digital games with a different goal than pure entertainment, and, thanks to their potential for engaging users, are particularly relevant for education and training, aiming at user motivation and behaviour improvement [1]. Many SGs have been developed to be used in class, or a safe area. Realistic 3D simulations, in fact, allow high fidelity training in environments and situations in which real-world training would be costly, dangerous or unfeasible. On the other hand, synergy with the Internet of Things (IoT) technologies [2] has a promising potential, especially for SGs related to field operations. The diffusion of low cost IoT devices, in fact, is enabling a new generation of games, that are implemented to work in the wild [2]. For example, in the road-traffic area, simulation-based games already support learning and practising [3], while new generation IoT-enabled games also aim to allow real-time interaction in the actual context of use [4] (e.g., driving game on smartphones equipped with inertial sensors [5][6]). Of course, given the frequently critical operation context, there are serious implications, in terms of distraction and citizen

privacy violation [7][8], that ought to be carefully taken into account in the design and deployment of such games. By no means should games become an instrument through which citizens are persuasively and “nicely” controlled.

Gamification is an approach that is being increasingly used to enhance the field experience with game elements [9][10], applying Game Theory [11]. On the other hand, Reality-Enhanced Serious Gaming (RESG) is a specialization of pervasive gaming in which games are fed with data collected from the field, also in real-time [12]. Thus, field users' performance becomes a key factor for the game itself, which is typically also played in the field or in the wild [13]. This aims to improve the user experience and strengthening the link with the actual activities to be improved.

In a RESG, field user performance should be estimated in real-time, in the order of a few seconds, with negligible computation latency assessment, in order to provide input to the game. Moreover, the user performance field assessment model should be easily understandable by game developers (that they may use the same model for different types of games), and be capable of giving coaching feedback to the player by supporting how to effectively identify different performance factors and how perhaps to actively manage these. Also, information sources for user assessment should be easily accessible, in order to allow a wide diffusion of the game. For this reason, we believe that the On-Board Diagnostics-II (OBD-II) interface [14] is a fundamental asset for an automotive game context.

We are exploring this area, aiming to improve automotive driver behaviour, as vehicles become ever more powerful IoT platforms [15]. Moreover, car drivers have significant margins to improve safety and reduce emissions, and these goals could be supported by well-designed SGs [16]. This paper reports on our work in the modelling stage of an IoT-enabled RESG. As an indicator of driver performance, we choose fuel consumption (FC), which is strongly influenced by driving styles [17][18][19], and can be quantified and validated. FC measurements, however, are not directly accessible through the OBD-II interface, which requires an elaboration of correlated signals. This reflects the general and common case in which user field performance is estimated by processing several information sources.

For estimating driver performance in terms of FC, we have taken the challenge by working with three OBD-II signals: throttle position sensor (TPS), revolutions per minute (RPM) and car speed, that are controllable by the driver, are comprehensible by him or her, while providing direct feedback to the driver or player [16]. The RESG – implemented on a smartphone or a raspberry pi connected via Bluetooth to the OBD-II – will exploit the real-time FC estimation values through a proper game mechanic (e.g., as energy available to the player avatar in a PassengerGame in [20]) and can provide real-time verbal recommendations to the driver on what actions he or she could do to improve his or her driving performance.

This paper focuses on analyzing two types of algorithms for driver performance modelling: fuzzy logic (FL) and random forest (RF). The former is a method of reasoning that resembles the human reasoning used in the development of human-like capabilities for Artificial Intelligence (AI). FL is a promising technique for driving style analysis [21]. It has the ability to deal with incomplete information [22][23], to distinguish between different performance factors by matching any set of input-output data, transferring human knowledge and expertise into a mathematical model by means of if-then rules [24], so to give coaching feedback to the driver about his or her performance. The latter is one of the ensemble learning methods, a powerful Machine Learning (ML) algorithm, that is quickly trainable, performs implicit feature selection by providing an indicator of feature importance [22][23].

In previous work [10], we implemented a FL model that was able to provide coaching advice to the driver, but we are now interested in exploring if a better quantitative estimation could be obtained with a ML technique. By combining the FL with RF as a ML technique, the game could provide coaching advice to the driver based on a FL model, and be integrated in the game (e.g., as an energy factor [20]) with a more accurate quantitative FC estimation obtained through a ML model, such as RF.

The experimental dataset for building, training and assessing our models, has been extracted from the enviroCar naturalistic historical open database [25]. We worked with data not calibrated for a specific car model, recorded in different driving environments, which made the work challenging and robust for real-world conditions, without the need for the driver or player to insert data about the vehicle nor train the system.

The remainder of the paper is organized as follows: Section 2 reviews the literature; Section 3 presents the data; Section 4 describes the FL model and the RF model; Section 5 presents our experiments to validate and assess the models; Conclusions and future works are drawn in Section 6.

II. BACKGROUND AND RELATED WORK

A growing number of IoT objects are being used as components in virtual or hybrid games [26]. In the specific automotive environment, in-car gaming is gaining relevance [27]. Several in-car game concept designs have been discussed in [28], also considering driving styles. In [29], an incentive-based mechanism was adopted to improve driver behaviour in managing traffic congestions. [30] evaluated the effects of gamification on driving, especially considering boredom. The Car-wings application by Nissan, represents

the FC status versus money spent and also provides a comparison amongst different drivers' performance [31]. The Car2Go application provides gamification features to support environmental friendly driving behaviour [32]. [33] discusses the positive and negative consequences of gamifying recreational bicycle riding.

[4] presented the TEAM RESG concept architecture, with three layers, distinguishing (1) sensing modules (e.g., the vehicular signals), from (2) game logic (e.g., virtual bank, snake & ladders, races), from (3) user interface (e.g., on a smartphone). Such architecture allows implementing various types of games, fed (e.g., directly as a score or indirectly as an energy factor, other game mechanics) by the seamless insertion of driver performance information (e.g., field user). The present work leverages this architecture, investigating algorithms able to estimate in real-time the performance of a user, that will be employed as a factor in an IoT driving SG.

Among ML techniques, an ensemble learning approach – the process of generating a team of ML models and then aggregating their results to obtain much powerful predictive performance, has gained a great interest nowadays. One of the most effective ensemble learning approaches is the random forest (RF) supervised technique, carrying prediction and classification problems, and it's also applicable for time series analysis. In RF, several decision or regression trees (a forest) are built and trained to empower predictive models with global more accurate and stable predictions, as a majority vote over all the trees in the forest for classification or average of outputs by different trees in the forest for regression [34][35]. Each tree is trained on a bootstrap sample, and optimal variables at each split of the tree's nodes are selected from a random subset of all available variables, which increases the purity of the node and results in the most homogeneous sub-nodes [36]. This strategy enables RF to be robust against overfitting and be held as an outstanding predictive model amongst many classification and regression tasks [37], including discriminant analysis, support vector machine for regression (SVR), and neural networks (NNs) [38]. RF methods are reasonably fast [38] and can be easily parallelised if more speed is required. In addition to prediction, RF is also considered as a dimensionality reduction and feature selection method, where it helps in the interpretation of important informative predictors used in predicting outcomes.

RF has been used in a myriad of domains, for example, [39] used RF to forecast droughts, and demonstrated that RF outperforms Autoregressive Integrated Moving Average. Also, RF has been applied in FC prediction. For instance, it was used to predict the FC of road vehicles based on on-board data in [40]. Another application is in [41], where FC estimation is used for heavy vehicles by combining the data from GPS, road, vehicle, and weather. The studies in [42] show that the RF technique produces a more accurate prediction with Mean-squared-error (MSE) equal to 0.001 (equivalent to a root mean square-error (RMSE) of 0.04) compared to both gradient boosting and NNs in FC modelling of a long distance public bus, given all available parameters as a time series. In [43], RF slightly outperforms SVR and NNs in FC prediction for trucks. This also includes extra-vehicle features, together with road condition data such as road geometry and the condition of the road infrastructure,

derived from fleet managers and road agency databases which affect significantly the fuel economy.

III. DATA

For our experiments, we relied on the enviroCar platform – a community-based open data collection platform for gathering pseudonymized naturalistic driving vehicular sensor data (cars are identified by ID numbers only) [25]. This community uses standard Bluetooth OBD-II adapters, that read information from the vehicle Controller Area Network or CAN bus. This information is sampled at regular time intervals (most of the tracks we used are recorded at a 5s sampling time) by an Android smartphone app, and delivered to the enviroCar server, together with GPS information for spatial-temporal analysis. Further information, like FC and Carbon dioxide emission estimation, is computed post-hoc and added on their server.

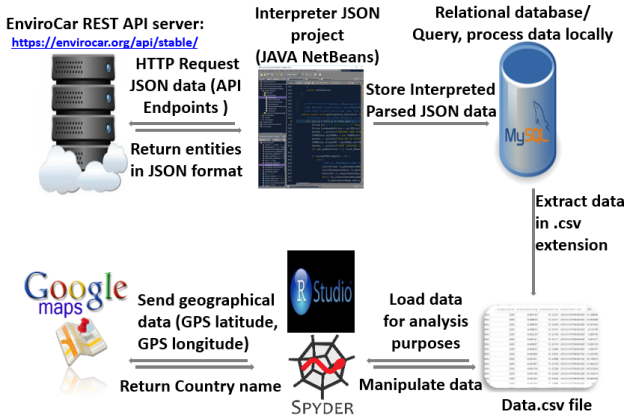


Fig. 1. Data system experimental architecture.

To build our dataset, we developed a software system that requests the data through a JSON (JavaScript Object Notation) interface, using enviroCar REST API via HTTP requests. Data are then stored in a local relational database for querying purposes (Fig. 1). In order to recognise the country where a track was recorded, we use the Google Maps API [44], provided by the “ggmap” R library [45], for the process of back (reverse) coding [46] a point location (latitude, longitude) into a readable address (country, locality, and route). For our analysis, we considered more than 1100 tracks, with 857,420 complete record measurements, that were recorded mostly in Germany in the period 2012-01-01 – 2016-06-15. From the enviroCar data, we focused on the three predictors (TPS, RPM and car speed) and the target (FC) to develop our FC models. In order to show the correlation between the target and the source signals, Fig. 2 shows the FC plots against TPS (a), RPM (b), and speed (c), for all the studied data. The Pearson Product Moment Correlation (PPMC) values between FC and our selected variables, are 0.82, 0.75 and 0.80 for RPM, car speed and TPS respectively (based on Fig. 4 in [16]).

The enviroCar community relies on the “Mass Air Flow” (MAF) sensor, which measures the amount of air that flows into the engine (the engine control unit uses it to determine how much fuel must be drawn through the engine cylinders). Consistently, our preliminary correlation analysis showed a PPMC value equal to 1 between FC and MAF (Fig. 4 in [16]). Although the MAF sensor is mandatory in the OBD-II standard, it is not supported by some vehicle types. Thus, the

community estimates it by utilising other parameters, namely temperature, air pressure, and engine speed [25].

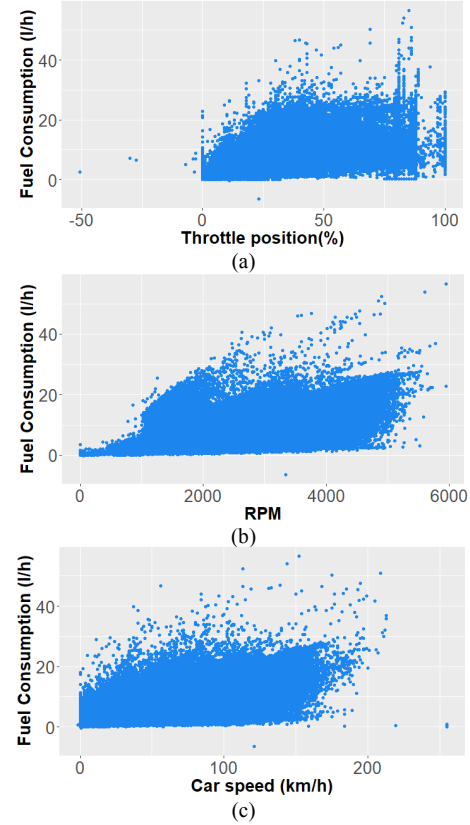


Fig. 2. Scatterplots from the enviroCar database: (1) fuel consumption based on throttle position; (2) fuel consumption based on RPM; (3) fuel consumption based on car speed.

We focus on gasoline engines, as enviroCar’s FC estimation provides the best accuracy for them [25]. EnviroCar estimation uses equation (1) [47], where MAF is measured in (grams/second) and AFR is the Air Fuel Ratio (Mass Ratio of Air to fuel), which is 14.7 for gasoline. MAF over AFR is thus in (grams/second). EnviroCar provides the FC in (liters/h), where gasoline has a 745 (grams/liter) density (equation (2)). A combustion is complete for a ratio of 14.7 kg of air per 1 kg of gasoline.

$$\text{Fuel consumption } \left(\frac{\text{g}}{\text{s}}\right) = \frac{\text{fuelweight}}{s} \left(\frac{\text{g}}{\text{s}}\right) = \frac{\text{MAF} \left(\frac{\text{g}}{\text{s}}\right)}{\text{AFR}} \quad (1)$$

$$\text{Fuel consumption } \left(\frac{\text{l}}{\text{h}}\right) = \frac{\frac{\text{MAF} \left(\frac{\text{g}}{\text{s}}\right)}{\text{AFR}}}{\text{Density of fuel } \left(\frac{\text{g}}{\text{l}}\right)} \left(\frac{\text{l}}{\text{s}}\right) * 3600 \left(\frac{\text{l}}{\text{h}}\right) \quad (2)$$

IV. THE FUZZY LOGIC (FL) AND RANDOM FOREST (RF) MODELS

This section presents the two developed FC models, exploiting the three statistically significant predictors (TPS, RPM and car speed).

A. Fuzzy Logic (FL)

The FL-based model was based on that of Mamdani [48]. The membership functions (MFs), reported in Fig. 3, were

defined based on literature expertise (for TPS, [49]), data analysis and trial and error (for the other signals). Table 1 provides the deduced 17 rules after studying and observing the 60 possible combinations of the MFs of the variables with the AND operator. All the rule weights are initially set to equal 1. Defuzzification is obtained through the centroid technique. As an example, based upon the 14th rule in Table 1, if a “High” FC is obtained because of a “High” RPM and a “High” TPS value, then the model can suggest to a player or driver to decelerate, or shift up a gear to save fuel.

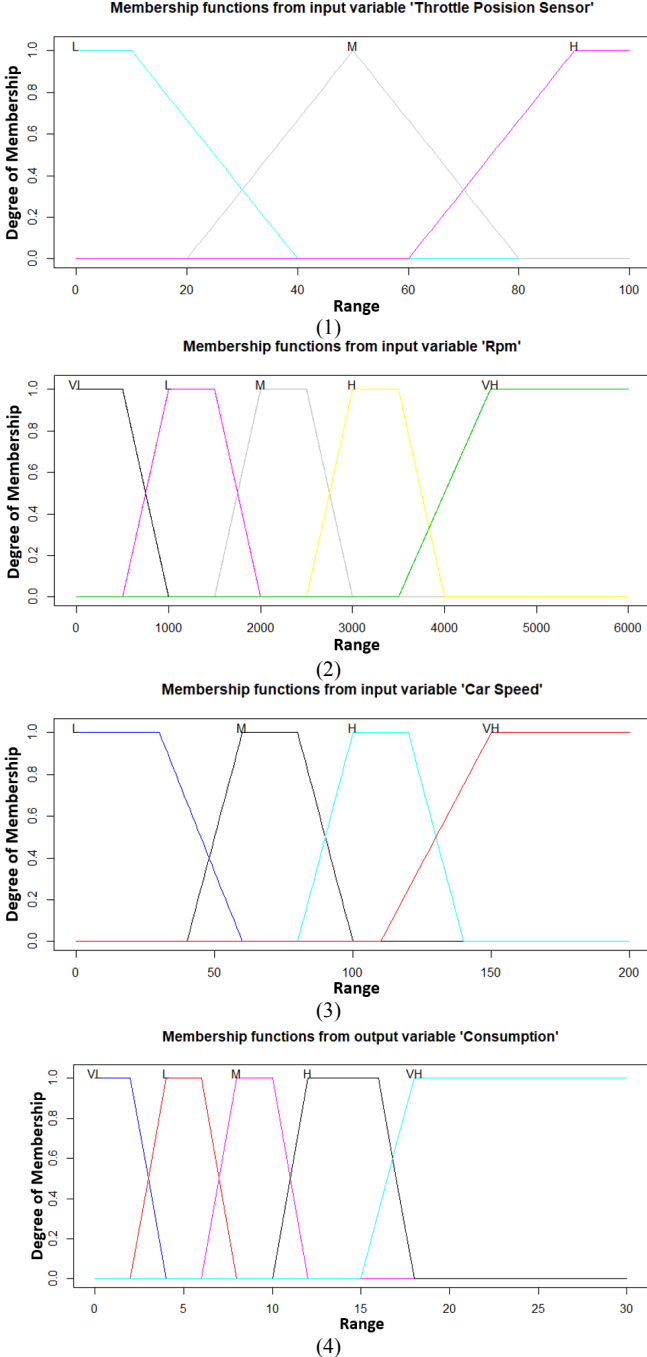


Fig. 3. Membership functions (MFs) for inputs and output. (1): Throttle position MFs. (2): RPM MFs. (3): Car speed MFs. VL is “Very Low”; L is “Low”; M is “Medium”; H is “High”; VH is “Very High”.

TABLE 1. FUZZY RULES.

FL Rules	
1	if RPM is VL then FC is VL
2	if RPM is L and TPS is L and Speed is (L or M or H) then FC is VL
3	if RPM is L and TPS is L and Speed is VH then FC is L
4	if RPM is L and TPS is M then FC is M
5	if RPM is L and TPS is H and Speed is (L or M) then FC is L
6	if RPM is L and TPS is H and Speed is H then FC is VH
7	if RPM is L and TPS is H and Speed is VH then FC is H
8	if RPM is M and TPS is L then FC is L
9	if RPM is M and TPS is M then FC is M
10	if RPM is M and TPS is H then FC is M
11	if RPM is H and TPS is L then FC is L
12	if RPM is H and TPS is M and Speed is L then FC is M
13	if RPM is H and TPS is M and Speed is (M or H or VH) then FC is H
14	if RPM is H and TPS is H then FC is H
15	if RPM is VH and TPS is L then FC is M
16	if RPM is VH and TPS is M then FC is H
17	if RPM is VH and TPS is H then FC is VH

B. Random Forest (RF)

For implementing the RF model, we used the “*RandomForestRegressor*” of the sklearn.ensemble python library [50]. The most important settings are the number of trees ($n_estimators$) in the forest and the number of features considered for splitting at each leaf node ($max_features$). For the developed RF, the following six hyperparameters have been adjusted,

- (1) $n_estimators$: number of trees to build in the forest.
- (2) $max_features$: max number of considered features for splitting a node.
- (3) max_depth : max number of levels in each decision tree.
- (4) $min_samples_split$: min number of data points placed in a node before the node is split.
- (5) $min_samples_leaf$: min number of data points allowed in a leaf node.
- (6) $bootstrap$: a method for sampling data points (with or without replacement).

Random search and grid search, have been used to determine the optimal parameters of the RF model. They have been implemented using “*RandomizedSearchCV*” [51] and “*GridSearchCV*” [52] respectively. Both methods are optimized by 10-fold cross-validation (CV) for reducing the chance of overfitting. First, we followed a random sampling approach during fitting, using the random search technique. We tried 60 different model settings (fitting 10-Fold CV for each of 6 candidates). At each iteration, the approach uses a random combination of the features to sample a wide range of values to end up with the best model. This helps in narrowing down the range for each hyperparameter values, that we define in a grid of their ranges as following $n_estimators$: [400,500,800,1000], $max_features$: [auto,sqrt], max_depth : [10,20,30,40,50,60,70,80,90,100,None], $min_samples_split$: [2,5,10,14], $min_samples_leaf$: [1,2,3,4,8] and $bootstrap$: [True, False]. The best model found by this approach is RF_1: [$n_estimators=800, max_features=sqrt,$

max_depth=90, min_samples_split=5, min_samples_leaf=4, bootstrap=False].

In a trial to further improve the results, we used a grid search approach to evaluate all combinations based upon the most promising ranges found in the random search: n_estimators:[700,800,900],max_features:sqrt,max_depth:[8,0,90,100],min_samples_split:[4,5,6],min_samples_leaf:[3,4,5] and bootstrap:True. The grid search fitting suggests the optimal model is RF_2: [n_estimators=800, max_features=sqrt, max_depth=100, min_samples_split=6, min_samples_leaf=3, bootstrap=False].

V. RESULTS AND DISCUSSIONS

The available dataset was divided into a training set (80% of the data) and a testing set (20%). Table 2 shows the accuracy of out-of-sample performance, the likely performance of the FL model and for both RF models RF_1 and RF_2 derived from a random search and a grid search respectively. Mean-squared-error (MSE) and squared correlation coefficient (R^2) statistical metrics have been used to assess the prediction performance.

TABLE 2. COMPARISON OF PERFORMANCE FOR THE FUZZY LOGIC (FL) AND RANDOM FOREST (RF) MODELS RF_1 AND RF_2.

Model	FL	RF_1	RF_2
MSE	4.745	1.539	1.506
R^2	0.650	0.893	0.896

We adopted to use the RF_2 model as suggested by the grid search technique, where the grid search's best found model is slightly better than the one found by the random search technique, with a slightly lower MSE (1.506 vs. 1.539) and a slightly higher R^2 (0.896 vs. 0.893).

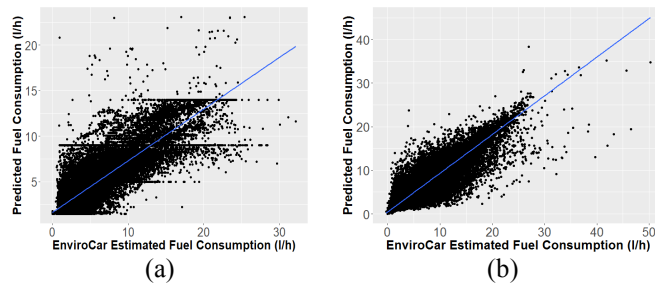


Fig. 4. Comparison of the model fit: (a) fuzzy logic, (b) random forest.

Comparing the two studied techniques, RF performs much better than FL, both in terms of R^2 (0.896 vs. 0.65) and MSE (1.506 vs. 4.74). Table 2 shows the fits of the FL model (a) and of the RF model (RF_2), for the testing set (171,484 samples). Fig. 5 visualizes the time evolution of FL (a) and RF (b) versus the actual enviroCar FC data in a sample track.

Considering the computation performance, which is crucial for gaming, we highlight that both FL and RF based models, can achieve a real-time execution, with latencies within a millisecond.

Observing the fuzzy rules to interpret the model (Table 1), we assumed that RPM is the most important FC predictor, followed by TPS and car speed [16]. For a more accurate picture, this study takes advantage of the RF feature

importance interpretation tool, by measuring the prediction strength of each variable.

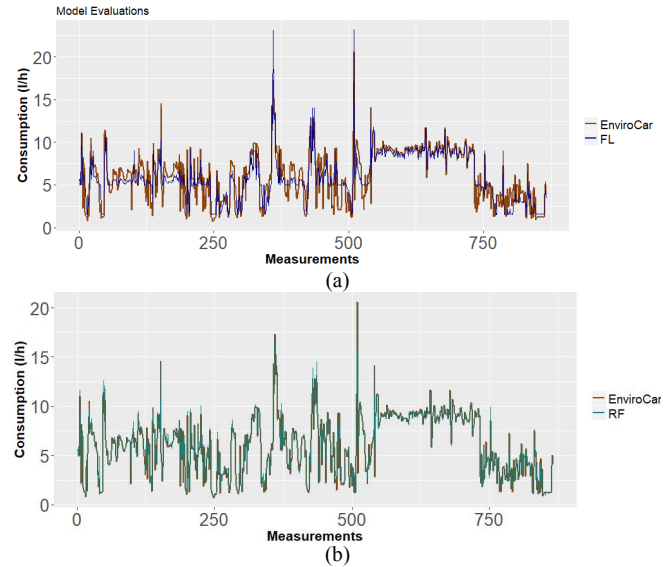


Fig. 5. Track data evaluation with the models fuzzy logic (a) and random forest (b).

For computing the feature importance in RF, we have applied two strategies. First, we relied on the most common mechanism – the mean impurity decrease (MID), default variable importance in scikit-learn [34]. Then we applied the permutation importance or Mean Decrease in Accuracy (MDA) recommended approach [53]. Upon [54], the permutation importance technique is broadly applicable for its reliability.

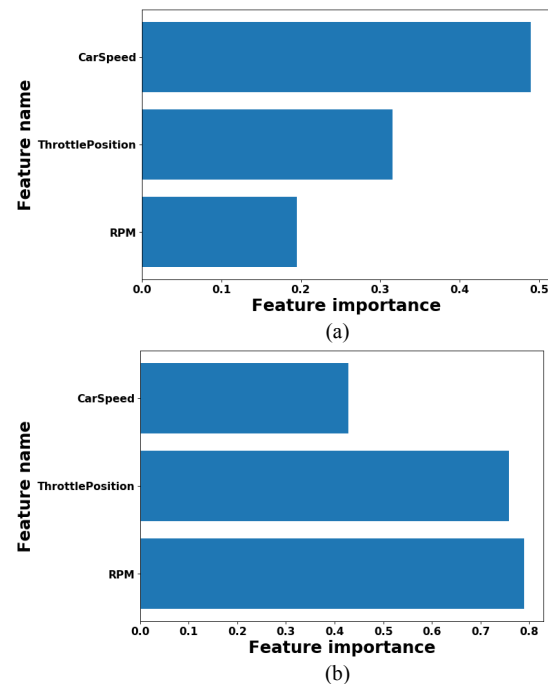


Fig. 6. Variable importance plots for the developed RF regressor. (a) Importance derived by Mean Impurity Decrease (MID). (b) Importance derived by permutation or Mean Decrease in Accuracy (MDA).

According to MID, car speed is the most influential feature of the variables in the FC model, followed by TPS and RPM respectively (Fig. 6 (a)). Comparing this with the results of our FL model [16], we observed that RPM is substituted by car speed as the strongest FC predictor.

According to the permutation approach (MDA) (Fig. 6 (b)), RPM is again the most important FC predictor, followed by TPS and car speed, confirming the main findings of our FL-based previous work [16].

This analysis is a key to a SG designer for a proper prioritisation of the predictors while supplying the player with instantaneous feedback to improve FC behaviour. Taking the previous example, based upon the 14th rule in Table 1, if a “High” FC is obtained because of a “High” RPM and a “High” TPS value, then the system should give priority to the gear shift message, and then to the deceleration suggestion.

VI. CONCLUSION AND FUTURE WORK

This paper reports our experience when exploring two types of algorithms for fuel consumption modelling for use in IoT-enabled driving games: fuzzy logic (FL) and random forest (RF). We took the challenge of defining and evaluating the models with naturalistic driving data, not calibrated for a specific car model, and recorded in different time slots in different days and in different driving environments and conditions. We spent a significant amount of time for the data analysis needed, to build the membership functions and the fuzzy rules for the FL model. Note also, the application of hyperparameter tuning techniques (random search and grid search) along with the 10-folds cross-validation for optimal RF model configurations was computationally expensive.

A FL model provides linguistic understandable feedback, concerning input to output links. It enables specifying the likely performance causes, based upon the deduced fuzzy rules, as presented in Table 1, resembling human reasoning, which is useful to support a player’s performance improvement, in a serious game. RF, on the other hand, is demonstrated to be able to predict fuel consumption much more accurately, with higher R^2 (0.896 vs. 0.65) and lower error (1.506 vs. 4.745) (Table 2), better capturing the trends in data. Moreover, RF provides a quantitative indicator of feature importance, according to which RPM is estimated to be the most informative signal, followed by throttle position and car speed.

These results suggest an opportunity to take advantage of both models, also considering that they both can support real-time execution, with latencies within a millisecond. We thus argue that the combination of the two models can provide valuable information usable in reality-enhanced serious games. FC estimations with good accuracy are needed as continuous sequential determinations of FC can be realised by a game design directly as an ongoing updated score, or as energy of the player, or to activate bonuses or maluses, or to facilitate reaching a higher gaming level, etc. (e.g., [13][20]). FL rules provide very good hints on the causes of the driver’s performance level, but RF allows achieving a much better quantitative estimation. Moreover, the RF feature importance tool gives important insights into the prioritization of the predictors, which in turn enhances the communication of the outcomes of the FL rules.

We finally argue that the RF and FL analysed techniques could be adopted for user field performance assessment, also in other application domains, besides vehicle driving, as they are able to process different source signals, for in the context

of the IoT-enabled Reality-Enhanced Serious Gaming concept.

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REFERENCES

- [1] F. Bellotti, R. Berta, and A. De Gloria, “Designing effective serious games: opportunities and challenges for research,” *International Journal of Emerging Technologies in Learning (IJET)*, vol. 5, 2010.
- [2] A. Al-Fuqaha, M. Guizani, M. Mohammadi, M. Aledhari, and M. Ayyash, “Internet of things: A survey on enabling technologies, protocols, and applications,” *IEEE Communications Surveys & Tutorials*, vol. 17, no. 4, pp. 2347–2376, 2015.
- [3] H. Prendinger, J. Oliveira, J. Catarino, M. Madruga, and R. Prada, “ICO2: a networked game for collecting large-scale eco-driving behavior data,” *IEEE Internet Computing*, vol. 18, no. 3, pp. 28–35, 2014.
- [4] F. Bellotti, S. Kopetzki, R. Berta, P. K. Paranthaman, G. R. Dange, P. Lytrivis, et al., “Team applications for collaborative road mobility,” *IEEE Transactions on Industrial Informatics*, 2018.
- [5] T. O. Oshin, S. Poslad, and Z. Zhang, “Energy-efficient real-time human mobility state classification using smartphones,” *IEEE Transactions on computers*, vol. 64, no. 6, pp. 1680–1693, 2015.
- [6] Z. Zhang and S. Poslad, “Fine-grained transportation mode recognition using mobile phones and foot force sensors,” in *International Conference on Mobile and Ubiquitous Systems: Computing, Networking, and Services*. Springer, 2012, pp. 103–114.
- [7] C. D. Jensen, S. Poslad, T. Dimitrakos. (Eds). *Proc. 2nd Int. Conf. on Trust Management, iTrust, Lecture Notes in Computer Science (LNCS)*, vol. 2995, Springer-Verlag, 2004.
- [8] S. Poslad, M. Hamdi, and H. Abie, “Adaptive security and privacy management for the internet of things (ASPI 2013),” in *Proceedings of the 2013 ACM conference on Pervasive and ubiquitous computing adjunct publication*, pp. 373–378.
- [9] F. Bellotti, R. Berta, A. De Gloria, E. Lavagnino, A. Antonaci, F. M. Dagnino, and M. Ott, “A gamified short course for promoting entrepreneurship among ICT engineering students,” *13th IEEE international conference on advanced learning technologies (ICALT)*, 2013, pp. 31–32.
- [10] S. Poslad, A. Ma, Z. Wang, and H. Mei, “Using a smart city IoT to incentivise and target shifts in mobility behaviour—Is it a piece of pie?” *Sensors*, vol. 15, no. 6, pp. 13 069–13 096, 2015.
- [11] H. Mei, S. Poslad, and S. Du, “A game-theory based incentive framework for an intelligent traffic system as part of a smart city initiative,” *Sensors*, vol. 17, no. 12, p. 2874, 2017.
- [12] J. D. Fijnheer and H. van Oostendorp, “Steps to design a household energy game,” *International Conference on Games and Learning Alliance*, 2015, pp. 12–22.
- [13] R. Drakoulis, F. Bellotti, I. Bakas, R. Berta, P. K. Paranthaman, G. R. Dange, P. Lytrivis, K. Pagle, A. De Gloria, and A. Amditis, “A Gamified Flexible Transportation Service for On-Demand Public Transport,” *IEEE Transactions on Intelligent Transportation Systems*, vol. 19, no. 3, pp. 921–933, 2018.
- [14] S. Godavarty, S. Broyles, and M. Parten, “Interfacing to the on-board diagnostic system,” in *52nd IEEE Vehicular Technology Conference (IEEE-VTS)*, vol. 4.
- [15] R. Kirk, “Cars of the future: the Internet of Things in the automotive industry,” *Network Security*, vol. 2015, no. 9, pp. 16–18, 2015.

- [16] R. Massoud, S. Poslad, F. Bellotti, R. Berta, K. Mehran, A. De Gloria, "A fuzzy logic module to estimate a driver's fuel consumption for reality-enhanced serious games," *International Journal of Serious Games*, vol. 5, no. 4, pp. 45–62, 2018.
- [17] J. N. Barkenbus, "Eco-driving: An overlooked climate change initiative," *Energy Policy*, vol. 38, no. 2, pp. 762–769, 2010.
- [18] P. H. Rettore, A. B. Campolina, L. A. Villas, and A. A. Loureiro, "A method of eco-driving based on intra-vehicular sensor data," *IEEE Symposium on Computers and Communications (ISCC)*, 2017, pp. 1122–1127.
- [19] R. Vaiana, T. Iuele, V. Astarita, M. V. Caruso, A. Tassitani, C. Zaffino, and V. P. Giofrè, "Driving behavior and traffic safety: an acceleration-based safety evaluation procedure for smartphones," *Modern Applied Science*, vol. 8, no. 1, p. 88, 2014.
- [20] G. R. Dange, P. K. Paranthaman, F. Bellotti, R. Berta, A. De Gloria, M. Raffero, and S. Neumeier, "Deployment of serious gaming approach for safe and sustainable mobility," *IEEE Intelligent Vehicles Symposium (IV)*, pp. 1222–1227.
- [21] G. A. M. Meiring and H. C. Myburgh, "A review of intelligent driving style analysis systems and related artificial intelligence algorithms," *Sensors*, vol. 15, no. 12, pp. 30653–30682, 2015.
- [22] F. Baig, M. W. Ashraf, M. Imran, Z. Ahmed, M. S. Khan, and M. A. Farooq, "Fuzzy Logic based Fuel Consumption System," *Bahria University Journal of Information & Communication Technology*, vol. 5, no. 1, p. 31, 2012.
- [23] Y. Bai and D. Wang, "Fundamentals of fuzzy logic control—fuzzy sets, fuzzy rules and defuzzifications," in *Advanced Fuzzy Logic Technologies in Industrial Applications*. Springer, 2006, pp. 17–36.
- [24] H. Nasution, H. Jamaluddin, and J. M. Syeriff, "Energy analysis for air conditioning system using fuzzy logic controller," *TELKOMNIKA (Telecommunication Computing Electronics and Control)*, vol. 9, no. 1, pp. 139–150, 2011.
- [25] A. Bröring, A. Remke, C. Stasch, C. Autermann, M. Rieke, and J. Möllers, "enviroCar: A Citizen Science Platform for Analyzing and Mapping Crowd-Sourced Car Sensor Data," *Transactions in GIS*, vol. 19, no. 3, pp. 362–376, 2015.
- [26] P. Coulton, D. Burnett, A. Gradinar, D. Gullick, and E. Murphy, "Game design in an internet of things," *Transactions of the Digital Games Research Association (ToDIGRA)*, vol. 1, no. 3, 2014.
- [27] P. Sundström, D. Wilfinger, A. Meschtscherjakov, M. Tscheligi, A. Schmidt, and O. Juhlin, "The car as an arena for gaming," *Proceedings of the 14th international conference on Human-computer interaction with mobile devices and services companion*, 2012, pp. 233–236.
- [28] S. Krome, S. P. Walz, S. Greuter, J. Holopainen, A. Gerlicher, and M. Schleeauf, "Exploring game ideas for stresslessness in the automotive domain," in *Proceedings of the 2014 Conference on Interactive Entertainment*, 2014, pp. 1–3.
- [29] R. McCall and V. Koenig, "Gaming concepts and incentives to change driver behavior," *11th Annual Mediterranean Ad Hoc Networking Workshop (Med-Hoc-Net)*, 2012, pp. 146–151.
- [30] F. Steinberger, R. Schroeter, and C. N. Watling, "From road distraction to safe driving: Evaluating the effects of boredom and gamification on driving behaviour, physiological arousal, and subjective experience," *Computers in Human Behavior*, vol. 75, pp. 713–726, 2017.
- [31] M. McFarland, "Why driving slowly and responsibly can actually be fun," *Washington Post*, 2014. Retrieved from https://www.washingtonpost.com/news/innovations/wp/2014/06/20/why-driving-slowly-and-responsibly-can-actually-be-fun/?noredirect=on&utm_term=.e25fb875d864 on 2018-06-23 on 2018-06-23.
- [32] J. Goldman, "Cargo2go'S Ecoscore And The Gamification," N.p. Retrieved from <https://jaygoldman.com/car2go-s-ecoscore-and-the-gamification-of-driving-d6415ea1045e> on 2017-01-17.
- [33] P. Barratt, "Healthy competition: A qualitative study investigating persuasive technologies and the gamification of cycling," *Health & place*, vol. 46, pp. 328–336, 2017.
- [34] L. Breiman, J. Friedman, C. J. Stone, R. A. Olshen, "Classification and regression trees. monterey, ca: Wadsworth. wadsworth statistics/probability series," 1984.
- [35] L. Breiman, "Random forests," *Machine learning*, vol. 45, no. 1, pp. 5–32, 2001.
- [36] A. Liaw, M. Wiener, et al., "Classification and regression by randomforest," *R news*, vol. 2, no. 3, pp. 18–22, 2002.
- [37] M. Herrera, L. Torgo, J. Izquierdo, and R. Pérez-García, "Predictive models for forecasting hourly urban water demand," *Journal of hydrology*, vol. 387, no. 1-2, pp. 141–150, 2010.
- [38] L. Rokach, "Ensemble-based classifiers," *Artificial Intelligence Review*, vol. 33, no. 1-2, pp. 1–39, 2010.
- [39] J. Chen, M. Li, and W. Wang, "Statistical uncertainty estimation using random forests and its application to drought forecast," *Mathematical Problems in Engineering*, vol. 2012, 2012.
- [40] H. Almér, "Machine learning and statistical analysis in fuel consumption prediction for heavy vehicles," Master's thesis, Kungliga Tekniska Högskolan (KTH) royal institute of technology, School of Computer Science and Communication (CSC), 2015.
- [41] J. Lindberg, "Fuel consumption prediction for heavy vehicles using machine learning on log data," Ph.D. dissertation, Master's thesis, KTH, School of Computer Science and Communications (CSC), 2014.
- [42] S. Wickramanayake, and H. D. Bandara, "Fuel consumption prediction of fleet vehicles using Machine Learning: A comparative study," *Moratuwa Engineering Research Conference (MERCon)*, 2016, pp. 90–95.
- [43] F. Perrotta, T. Parry, and L. C. Neves, "Application of machine learning for fuel consumption modelling of trucks," in *IEEE International Conference on Big Data (Big Data)*, 2017, pp. 3810–3815.
- [44] Google Maps Platform. Retrieved from <https://cloud.google.com/maps-platform/>, on 2017-02-10.
- [45] Reverse geocode in R with ggmap package. Retrieved from <https://www.rdocumentation.org/packages/ggmap/versions/2.6.1/topic/s/revgeocode>, on 2017-04-07.
- [46] Google Maps reverse geocoding. Retrieved from <https://developers.google.com/maps/documentation/geocoding/start>, on 2017-04-10.
- [47] A. Faiz, C. S. Weaver, and M. P. Walsh, *Air pollution from motor vehicles: standards and technologies for controlling emissions*. The World Bank, 1996.
- [48] E. H. Mamdani, "Application of fuzzy algorithms for control of simple dynamic plant," in *Proceedings of the institution of electrical engineers*, vol. 121, no. 12. IET, 1974, pp. 1585–1588.
- [49] W. Wang, J. Xi, and X. Li, "Statistical pattern recognition for driving styles based on Bayesian probability and kernel density estimation," *arXiv preprint arXiv:1606.01284*, 2016.
- [50] RandomForestRegressor package. Retrieved from, <http://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestRegressor.html>, on 2018-11-02.
- [51] RandomizedSearchCV method of sklearn.model_selection. Retrieved from https://scikit-learn.org/stable/modules/generated/sklearn.model_selection.RandomizedSearchCV.html, on 2018-11-10.
- [52] GridSearchCV method of sklearn.model_selection. Retrieved from https://scikit-learn.org/stable/modules/generated/sklearn.model_selection.GridSearchCV.html, on 2018-11-10.
- [53] L. Breiman, "Random forests leo breiman and adele cutler," *Random Forests-Classification Description*, 2015. Retrieved from https://www.stat.berkeley.edu/~breiman/RandomForests/cc_home.htm, on 2018-10-16.
- [54] T. Parr, K. Turgutlu, C. Csizsar, and J. Howard, "Beware Default Random Forest Importances," 2018 Retrieved from <https://explained.ai/rf-importance/index.html>, on 2018-11-12.